SmartUp Platform — Daily Testing Plan

Period: 29 April 2026 – 15 May 2026 **Goal:** Fully bug-free, production-ready web platform by 15 May 2026 **Testers:** Dev Team, Staff, Offline Students **Devices:** Android, iOS, Desktop (Chrome, Safari, Firefox, Edge, Internet Explorer)

Test Accounts — `Smartup@2026`

Role	Email	Name
Owner	`smartuplearningventures@gmail.com`	Admin Owner
Owner	`owner@smartupedu.in`	—
HR	`hr@smartupedu.in`	—
Academic Operator	`chinjurp333@gmail.com`	Academic Operator
Batch Coordinator	`Academiccoordinator1SUO@gamil.com`	Batch Coordinator
Teacher (Math)	`salihr@smartupedu.in`	Salih R
Teacher (Physics)	`salehl@smartupedu.in`	Saleh L
Teacher (Chemistry)	`bilal@smartupedu.in`	Bilal
Teacher (Biology)	`shafivk@smartupedu.in`	Shafi VK

Operational Flow (How the Platform Is Actually Used)

```
STEP 1 — Platform Configuration (one-time, pre-launch)
  Owner configures fee structures
  → AO configures academic settings

STEP 2 — Student Acquisition
  Sales gets lead → demo (guest session if live class nearby, else demo link)
  → prospect observes/attends demo → sales follows up → lead converts
  [NOTE: "Add Guest to Session" feature pending — see
ENROLLMENT_FLOW_CHANGES.md]

STEP 3 — Enrollment (after lead closes)
  Sales sends enrollment link via CRM
  → student self-enrolls: fills details + pays via Razorpay
  → account created, credits added
  [Special case only: AO manually adds student — offline payment, exceptions]

STEP 4 — Batch Setup (after student is enrolled)
  AO creates batch for the enrolled student's grade/type (if not already
existing)
  → AO assigns coordinator to batch
  → AO assigns enrolled student to batch → join tokens created → welcome
email sent
```

```
STEP 5 — Academic Calendar & Scheduling
  AO/Teacher sets up academic calendar → maps curriculum topics
  → AO schedules sessions (single + recurring) → assigns teachers
  → topics linked to sessions from calendar
  [NOTE: BC session scheduling UI not yet implemented — see
CODE_CHANGES_NEEDED.md]

STEP 5 — Pre-Class
  Session reminder sent 15 min before → student gets WhatsApp/email
  → student opens join link → pre-join lobby (device check)
  → BC online (heartbeat active) → teacher enters room

STEP 6 — Live Class
  Teacher goes live (BC approves) → students join → class runs
  → controls (mic, cam, screen share, whiteboard) → chat, raise hand
  → session exam (optional) → homework assigned → teacher ends class

STEP 7 — Post-Class
  Attendance recorded → feedback collected → exam auto-graded
  → result emails sent → monitoring report generated

STEP 8 — Ongoing Operations
  New student added mid-batch → tokens backfilled → joins next session
  Teacher leave → substitute assigned
  Student non-payment → gate blocks → payment reminder sent
  Parent monitors via parent portal
  AO/ghost monitors live classes silently
```

Workflow Suggestions & Pending Improvements

> Identified during planning and codebase study. Prioritised by launch impact. Implementation details in `docs/CODE_CHANGES_NEEDED.md`.

#	Suggestion	Priority	Reason
1	**Batch optional in ManualEnrollModal** — Step 2 currently requires batch selection. Allow AO to enroll a student without assigning a batch immediately (assign later).	🔴	AO may add student before a suitable batch exists
2	**Enrollment link → auto-assign batch** — When AO creates an enrollment link, allow linking a target batch. After student pays, auto-assign and send join links without AO intervention.	🔴	Removes the manual "assign to batch" step; faster student onboarding
3	**BC go-live timeout** — If BC does not respond to go-live request within 2 min, allow teacher to go live with AO notification. Currently blocks indefinitely if BC is offline.	🔴	Classes stall when BC is briefly unavailable
4	**BC session scheduling (pending)** — Implement BC session scheduling UI (see `docs/CODE_CHANGES_NEEDED.md`). Day-to-day scheduling should be BC's job, not AO's.	🔴	AO is overloaded; BC is the right role for daily session management

#	Suggestion	Priority	Reason
5	**Session reminder — configurable window** — Allow per-batch reminder timing (15 min / 30 min / 1 hour). Currently hardcoded at 15 min for all batches.	🟡	Evening classes and long sessions need earlier reminders
6	**Auto-link academic topic to session** — When scheduling a session, suggest the next unfinished topic from the academic calendar for auto-linking. Currently fully manual.	🟡	Reduces AO workload; keeps curriculum tracking accurate automatically
7	**Session gate prompt — add context** — When student is blocked, show: batch name, how many sessions, exact amount, and fee breakdown. Currently shows only total amount.	🟡	Students don't understand why they're blocked or how much to pay
8	**Enrollment link expiry — configurable** — Currently fixed at 7 days. Allow AO to set custom expiry per link (24h for urgent enrollments, 30d for campaigns).	🟡	Different sales cycles need different expiry windows
9	**WhatsApp join link — shortened URL** — Current join links are long tokens. Consider a URL shortener or cleaner format for better mobile display in WA.	🟢	Long URLs wrap awkwardly in some WhatsApp clients
10	**Parent notification preferences** — Allow parents to opt in/out of specific notification types (exam results, attendance alerts, fee reminders). Currently all-or-nothing.	🟢	Reduces notification fatigue while keeping critical alerts

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Testing Phases

Phase	Label	Meaning
🔴 Red	Critical	Blocks usage — stop everything, fix immediately
🟡 Yellow	Important	Degraded experience — fix before launch
🟢 Green	Polish	Minor UX — fix before launch, lower urgency

Batch Types

Batch	Students	Behaviour
1:1	1	Private — student can publish mic/cam
1:3	3	Small group — students can publish
1:15	15	Medium group — students can publish
1:30	30	Large group — students can publish
1:Many / Lecture	50+	Lecture mode — mic/cam disabled for students

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Browsers & Devices

Browser	Desktop	Android	iOS
Chrome	✓	✓	✓
Safari	✓ Mac	—	✓
Firefox	✓	✓	✓
Edge	✓	✓	—
Internet Explorer	✓ Win	—	—
Samsung Internet	—	✓	—

> IE has no WebRTC — must show a graceful "browser not supported" page, never a blank crash.

Role Responsibilities

Role	Key Responsibilities
Academic Operator (AO)	Creates batches · assigns coordinators · **schedules sessions** (current implementation) · adds students manually · verifies CRM students · manages payments · ghost monitors · full reports
Batch Coordinator (BC)	**Session scheduling (pending implementation)** · assigns teachers to sessions · live monitoring · attendance · cancellations · batch-level reports (assigned batches only)
Teacher	Joins room · teaches · controls (mic/cam/screen/whiteboard) · attendance · homework · exams
Student	Joins via link · attends class · chat · raise hand · exams · homework submission
Owner	Complete platform monitoring · view all batches/sessions/users/reports · permission management for all roles · system config (read-only)
HR	User management · payroll · teacher leave approvals
Sales	Leads · demo link sending · pipeline
Parent	View child attendance/fees/exams · file complaints
Ghost	Silent observer — invisible to all participants

> **AO currently schedules sessions.** BC session scheduling is planned but not yet implemented in code. **BC cannot create batches or access finance.** AO handles these. **BC sees only assigned batches**, not all platform batches. See `docs/CODE_CHANGES_NEEDED.md` for the BC session scheduling implementation plan.

Bug Report Format

```
Bug ID    : BUG-[MMDD]-[###]    e.g. BUG-0501-001
Feature   : e.g. Session Gate
Batch Type: 1:1 / 1:3 / 1:15 / 1:30 / 1:many
```

```
Device    : Android / iOS / Desktop
Browser   : Chrome / Safari / Firefox / Edge / IE
Role      : Student / Teacher / BC / AO / Owner / HR / Sales / Parent / Ghost
Steps     : 1. ...  2. ...  3. ...
Expected  : What should happen
Actual    : What happened
Phase     : 🔴 / 🟡 / 🟢
Status    : Open / In Progress / Fixed / Verified
```

DAILY TEST SCHEDULE

Day 1 — 29 April 2026

Theme: FULL PLATFORM FIRST RUN — Enroll Students → Create Batches → Schedule → Run Live Classes → All Features

> End-to-end first-run test. 30 students manually enrolled by AO, batches built, all batch types run simultaneously. Three phases: **Setup** (students + batches + schedule) → **Live** (run all classes) → **Scenarios** (edge cases mid-session and post-session).

PHASE A — Student Enrollment (Manual, by AO)

> AO manually enrolls 30 test students using the 3-step Manual Enrollment form (Students tab → Manual Enroll button). Form steps: Step 1 = Student Info · Step 2 = Parent & Batch · Step 3 = Payment. Batches must exist BEFORE enrolling — create batches first (Phase B), then come back to enroll students into them. **Actual order on Day 1:** Create batches (Phase B) → enroll students into them (Phase A).

#	Scenario	Role	Steps	Expected	Phase
A1	AO login on Desktop Chrome	AO	Login → check dashboard	Dashboard loads, all tabs visible (Batches, Students, Sessions, Finance, Reports)	🔴
A2	Open Manual Enrollment modal	AO	Students tab → Manual Enroll button	3-step modal opens: Step 1 "Student Info" shown	🔴
A3	Step 1 — fill student info	AO	Name, Email, Phone, Grade=Class 10, Board=CBSE, Region=India, Category=A, DOB, Password	All fields accept input, validation on invalid email shows error	🔴
A4	Step 1 — missing name → blocked	AO	Submit Step 1 with empty name	Error: "Student name is required" — cannot proceed	🔴

#	Scenario	Role	Steps	Expected	Phase
A5	Step 1 — invalid email → blocked	AO	Enter "notanemail" in email field	Error: "Invalid email address"	🔴
A6	Step 2 — parent info (optional)	AO	Parent Name, Parent Email, Parent Phone, Parent Password	Fields accept input; leaving blank is allowed	🔴
A7	Step 2 — batch dropdown loads	AO	Batch selector on Step 2	All active batches listed with name, grade, board, student count/max	🔴
A8	Step 2 — select 1:1 batch	AO	Choose 1:1 batch from dropdown	Batch info card shows: Type=1:1, Students=0/1	🔴
A9	Step 2 — no batch selected → blocked	AO	Click Next without selecting batch	Error: "Please select a batch"	🔴
A10	Step 3 — payment mode: None	AO	Leave "No payment now" selected → Submit	Enrollment succeeds, no invoice generated	🔴
A11	Step 3 — payment mode: Full, Cash	AO	Select "Full payment", method=Cash, amount=₹5000 → Submit	Enrollment succeeds, invoice + receipt generated	🔴
A12	Step 3 — payment mode: Advance, UPI	AO	Advance, UPI, amount=₹2000, transaction ref	Advance payment recorded, partial credit	🔴
A13	Success screen shown	AO	After submit	"Student Enrolled!" screen shows: student email, batch name, invoice number, receipt number, amount paid	🔴
A14	"Enroll Another" resets form	AO	Click "Enroll Another" on success screen	Modal resets to Step 1, ready for next student	🔴
A15	Enroll 1 student into 1:1 batch	AO	Manual enroll → 1:1 QA batch → no payment	Student S1 enrolled, welcome email sent	🔴

#	Scenario	Role	Steps	Expected	Phase
A16	Enroll 3 students into 1:3 batch	AO	Repeat x3 → 1:3 QA batch	Students S2, S3, S4 enrolled	🔴
A17	Enroll 15 students into 1:15 batch	AO	Repeat x15 → 1:15 QA batch	S5–S19 enrolled, batch fills up	🔴
A18	Enroll 10 students into 1:30 batch	AO	Repeat x10 → 1:30 QA batch	S20–S29 enrolled	🔴
A19	Enroll 1 student into lecture batch	AO	Repeat x1 → lecture QA batch	S30 enrolled	🔴
A20	Total: 30 students enrolled	AO	View Students tab	All 30 students visible in list with correct batch assignments	🔴
A21	Duplicate email rejected	AO	Try to enroll S1's email again	Error: "Email already registered" — not a silent duplicate	🔴
A22	1:1 batch full — cannot add more	AO	Enroll 2nd student into 1:1 batch	Error: batch is full / max students reached	🔴
A23	Student welcome email received	Student	Check S1 email inbox	Welcome email with login credentials and dashboard link received	🔴
A24	Parent welcome email received	Parent	Check parent email (if provided)	Parent gets login credentials	🟡
A25	Enrolled student can login	Student	Use credentials from welcome email → login	Dashboard loads, student sees their batch and upcoming sessions	🔴

PHASE B — Batch Creation & Assignment

> AO creates all batch types for QA testing. Do this BEFORE Phase A so batches appear in the enrollment dropdown.

#	Scenario	Role	Steps	Expected	Phase
B1	AO creates batch — 1:1	AO	Batches → New → Type=1:1, Name="QA 1:1 Math", Grade=Class 10, Board=CBSE, Subject=Math	Batch saved, max_students=1, appears in list	🔴

#	Scenario	Role	Steps	Expected	Phase
B2	AO creates batch — 1:3	AO	Type=1:3, Name="QA 1:3 Math", Grade=10, Board=CBSE, Subject=Math	max_students=3	🔴
B3	AO creates batch — 1:15	AO	Type=1:15, Name="QA 1:15 Science", Grade=10, Board=CBSE, Subject=Science	max_students=15	🔴
B4	AO creates batch — 1:30	AO	Type=1:30, Name="QA 1:30 English", Grade=10, Board=CBSE, Subject=English	max_students=30	🔴
B5	AO creates batch — lecture	AO	Type=lecture, Name="QA Lecture Physics", Grade=11, Board=CBSE, Subject=Physics	max_students=50+	🔴
B6	AO assigns BC to all batches	AO	Each batch → Edit → Coordinator = official4tishnu@gmail.com	BC sees all 5 QA batches in their dashboard	🔴
B7	AO assigns primary teacher to each batch	AO	Each batch → Teachers → Add → abcdqrst404@gmail.com, subject=match batch subject	Teacher assigned as primary	🔴
B8	BC login — sees only assigned batches	BC	Login → Batches	Only the 5 QA batches visible — no other platform batches	🔴
B9	BC cannot create batch	BC	Try Batches → New	No "New Batch" button or option available	🔴
B10	AO views all batches platform-wide	AO	Batches list	ALL batches across platform shown, not filtered by BC	🔴

PHASE C — Session Scheduling

> AO schedules one session per batch, all at the same time (e.g., today + 30 min from now), to run simultaneously.

#	Scenario	Role	Steps	Expected	Phase
C1	AO schedules session — 1:1 batch	AO	Sessions → New → QA 1:1 batch, Date=today, Time=T+30min, Duration=60min, Subject=Math, Teacher=abcdqrst404@gmail.com	Session created, status=scheduled	🔴

#	Scenario	Role	Steps	Expected	Phase
C2	AO schedules session — 1:3 batch	AO	Same params, QA 1:3 batch, same time	Session created	🔴
C3	AO schedules session — 1:15 batch	AO	Same, QA 1:15 batch	Session created	🔴
C4	AO schedules session — 1:30 batch	AO	Same, QA 1:30 batch	Session created	🔴
C5	AO schedules session — lecture batch	AO	Same, QA Lecture batch	Session created	🔴
C6	Students receive join link emails	Students	After scheduling	All enrolled students receive session join link via email (+ WA if configured)	🔴
C7	Session reminder at 15 min before	Auto	Wait until T+15 min	WhatsApp + email reminder sent to all students and parents	🔴
C8	AO changes teacher on scheduled session	AO	Sessions → QA 1:3 session → Edit → change teacher	Teacher updated, new teacher notified, old teacher loses session	🔴
C9	AO reschedules session time	AO	Sessions → QA 1:1 session → Edit → change time by +5 min	Session time updated, students re-notified	🟡
C10	Session without teacher — warning shown	AO	Create new session without assigning teacher	Warning: "teacher not assigned" shown	🟡

PHASE D — Live Class (All 5 Batches Running Simultaneously)

> Teacher, students, BC, and AO all join at the same time. Test all controls, connectivity, and features.

#	Scenario	Role	Steps	Expected	Phase
D1	BC approves teacher go-live — 1:3 batch	BC	BC logs in, opens go-live approval panel	Go-live request appears when teacher tries to start	🔴
D2	Teacher enters room — 1:3 batch	Teacher	Session → Join → pre-join lobby → enter room	Room loads, teacher's own video tile shows, waiting for BC approval	🔴
D3	BC approves → class goes live	BC	BC → Approve go-live	Room status changes to live, students can now join	🔴
D4	Students join — 1:3 batch (3 students)	S2, S3, S4	Open join link → lobby → enter room	All 3 students appear in room, attendance recording starts	🔴
D5	Students join — 1:15 batch (15 students)	S5–S19	Open join links → enter room	All 15 tiles render correctly, no layout crash	🔴
D6	Students join — 1:30 batch (10 students)	S20–S29	Open join links → enter room	All 10 tiles render, grid adapts	🔴
D7	Student joins — 1:1 batch	S1	Open join link → enter room	1:1 room with teacher only — correct	🔴
D8	Student joins — lecture batch	S30	Open join link → enter room	Lecture mode: mic/cam controls disabled for student	🔴
D9	Teacher mic on/off	Teacher	Toggle mic button	Audio cuts for all students, icon reflects state	🔴
D10	Teacher camera on/off	Teacher	Toggle camera button	Video freezes/resumes for all students	🔴
D11	Teacher screen share — Desktop Chrome	Teacher	Screen share button → select window	Students see teacher's screen instead of camera	🔴

#	Scenario	Role	Steps	Expected	Phase
D12	Screen share stops	Teacher	Stop sharing	Teacher camera resumes	🔴
D13	Student mic on/off — 1:3 batch	Student S2	Toggle mic	Audio heard by teacher + other students in 1:3	🔴
D14	Student camera on/off — 1:3 batch	Student S3	Toggle camera	Video shown/hidden correctly	🔴
D15	Lecture mode: student cannot unmute	Student S30	Try to unmute in lecture batch	Mic button disabled or blocked — cannot publish audio	🔴
D16	Raise hand	Student	Raise Hand button	Teacher sees raised hand indicator on student tile	🔴
D17	Teacher lowers hand	Teacher	Click lower hand on student	Hand indicator clears	🔴
D18	Chat — student sends message	Student	Type in chat → send	Message appears for teacher and all students	🔴
D19	Chat — teacher sends message	Teacher	Type and send	All students receive message	🔴
D20	Teacher whiteboard / annotation	Teacher	Open whiteboard or annotation tool	Drawing visible to all students in real time	🔴
D20a	Mute individual student	Teacher	Right-click S3 tile → Mute	S3's audio cuts; mute icon updates on tile for all viewers; sound effect plays	🔴
D20b	Unmute individual student	Teacher	Right-click S3 tile → Unmute	S3's mic reactivates; icon updates	🟡
D20c	Remove student from room	Teacher	Controls → Remove S4 → confirm	S4 disconnected immediately; sees "You have been removed" message	🔴

#	Scenario	Role	Steps	Expected	Phase
D20d	Removed student blocked from rejoin	S4	S4 opens join link again	"Removed from session" — access denied, not a blank crash	🔴
D20e	Teacher announcement bar	Teacher	Toolbar → Announcement → type "Check page 42" → Send	Announcement banner overlays all student views; auto-dismisses after timeout	🟡
D20f	Spotlight student	Teacher	Click S2 tile → Spotlight	S2's tile enlarges to main stage; all others move to strip	🟡
D20g	Student submits doubt	Student S5	Doubts panel → type question → Submit	Question appears in teacher's doubts list in sidebar	🟡
D20h	Teacher responds to doubt	Teacher	Doubts → S5's question → type reply → Send	S5 sees teacher's reply in their doubts panel	🟡
D20i	Tab switch detection — teacher notified	Student S6	S6 switches browser tab during session	Teacher sees tab-switch violation alert with S6's name + sound cue	🟡
D20j	Contact violation — phone number in chat	Student S7	S7 types "+91 9876543210" in chat	Message blocked or flagged; teacher notified; warning shown to S7	🟡
D20k	Contact violation — social handle	Student S8	S8 types "@username" in chat	Blocked or flagged; teacher notified	🟡
D20l	Recording start	Teacher	Controls → Record (allowRecording=true)	REC badge appears for teacher; recording begins server-side	🟡

#	Scenario	Role	Steps	Expected	Phase
D20m	Recording stop	Teacher	Click Stop Recording	REC badge gone; recording URL saved; accessible later	🟡
D20n	Virtual background — blur	Teacher	Settings → Background → Blur	Teacher's background blurred; students see blurred background	🟢
D20o	Virtual background — custom image	Teacher	Settings → Background → Select image	Custom image shown behind teacher	🟢
D20p	Virtual background — disable	Teacher	Settings → Background → Off	Normal camera feed restored	🟢
D20q	Media request — lecture student requests mic	S30	Lecture batch → Request Mic button → submit	Teacher hears sfxMediaRequest; media request notification appears in panel	🟡
D20r	Teacher approves media request	Teacher	Media Requests panel → Approve S30	S30 can speak; all other lecture students remain muted	🟡
D20s	Teacher denies media request	Teacher	Deny S30	S30 stays muted; denial notification shown to S30	🟡
D20t	Teacher tablet joins (teacher_screen role)	Teacher	2nd device → login as teacher_screen → join session	Tablet appears as secondary screen-share source in room; students see it	🟡
D21	AO ghost joins 1:15 session	AO	Open session as ghost / AO role	AO invisible in participant list, can see all tiles and hear audio	🔴
D22	BC monitoring view — 1:15 batch	BC	CoordinatorLiveView → 1:15 room	All student tiles + attention states shown	🔴
D23	Student count badge —	Teacher	Check participant count display	Shows correct number present	🔴

#	Scenario	Role	Steps	Expected	Phase
	correct			(not counting ghost)	
D24	Teacher pushes session exam	Teacher	Exam tab → Create MCQ → Push to students	All students get exam overlay in their view	🔴
D25	Students submit exam	All students	Answer MCQs → submit before timer	All submissions recorded, auto-graded	🔴
D26	Teacher assigns homework	Teacher	Homework tab → New → title, description → assign	Homework visible in student homework list	🟡
D27	Teacher ends class — 1:3 batch	Teacher	End Session button	All participants disconnected, session status → ended	🔴
D28	Attendance recorded after end	AO/BC	Check attendance report	Join time, leave time, duration correct for all students	🔴
D29	Exam results emailed	Students + Parents	After session ends	Each student gets exam result email	🔴
D30	Post-session feedback dialog	Student	Immediately after end	Feedback/rating dialog appears for student	🟡
D31	Go-live skip toggle (bypass BC approval)	AO	AO → Batch settings → `go_live_skip_coordinator = true` → teacher goes live	Goes live immediately without BC request — no approval dialog	🔴
D32	Go-live before schedule (early start)	AO	AO → `allow_go_live_before_schedule = true` → teacher goes live 10 min early	Room goes live before scheduled time; no error	🟡
D33	Session runs past scheduled end (overtime)	Teacher	Continue class past 60-min mark	Overtime timer shows; session stays live; no auto-disconnect	🟡
D34	Teacher requests early	Teacher + BC	Teacher → End Early → BC sees approval dialog → Approves	Class ends; all disconnected; attendance	🟡

#	Scenario	Role	Steps	Expected	Phase
	end → BC approves			closed at actual end time	
D35	Teacher requests early end → BC denies	Teacher + BC	Teacher → End Early → BC denies	Class continues; teacher sees "denied" notification	🟡

PHASE E — Additional Scenarios (Mid/Post Session)

> Edge cases, role boundaries, and batch management scenarios.

#	Scenario	Role	Steps	Expected	Phase
E1	Add student to batch mid-series	AO	Batch → Add existing student (S20) to 1:3 batch (already in 1:30)	Student added to second batch, backfilled tokens for upcoming sessions	🔴
E2	New student backfill — join link works	S20	Use backfilled join link	Enters room successfully for the next session	🔴
E3	Change teacher after session is live	AO	Session (still live) → Edit → assign different teacher	Not allowed mid-session OR updated and takes effect next session	🟡
E4	AO removes student from batch	AO	Batch → Students → remove S4	S4 removed, future join tokens invalidated	🟡
E5	Removed student cannot join	S4	Try old join link	"Access denied" or "Session not found" — not a crash	🟡
E6	BC cannot access finance	BC	Try to navigate to fees/invoices	Not in nav, access denied if URL typed directly	🔴
E7	BC changes coordinator — not possible	BC	Try to reassign coordinator of their own batch	No option available	🔴
E8	Student joins wrong batch link	Student	Use another student's join link	Access denied — not the assigned student	🟡
E9	AO deactivates a batch	AO	Batch → set inactive	Batch greyed out, cannot schedule new sessions	🟡
E10	AO reactivates batch	AO	Batch → set active	Schedulable again	🟡
E11	Multiple teachers per subject	AO	1:15 batch → Teachers → add Teacher B for Science	Both teachers listed; AO can assign either per	🟡

#	Scenario	Role	Steps	Expected	Phase
			(same subject as Teacher A)	session	
E12	AO changes teacher of scheduled session	AO	Session → Edit → change teacher from A to B	Session updated, Teacher B notified, Teacher A loses session	🔴
E13	Teacher sees updated session	Teacher B	Dashboard → My Sessions	New session appears	🔴
E14	Teacher A no longer sees session	Teacher A	Dashboard → My Sessions	Session removed from Teacher A's list	🔴
E15	Student joins — Android Chrome	Student	Open join link on Android Chrome	Room loads, mic and camera work	🔴
E16	Student joins — iOS Safari	Student	Open join link on iPhone Safari	Room loads, permissions prompt shown	🔴
E17	Student joins — Desktop Firefox	Student	Open join link on Firefox	Room loads, all controls work	🔴
E18	Student joins — Desktop Edge	Student	Open join link on Edge	Room loads	🟡
E19	Pre-join lobby — mic denied	Student	Browser popup → deny mic → enter	"Mic denied" warning shown, can still enter room	🔴
E20	Pre-join lobby — no camera	Student	Device has no camera	"No camera found" shown, can still enter	🔴
E21	Expired join link	Student	Use join link after session ended	"Session has ended" — not a crash	🔴
E22	Invalid join link	Student	Open random /join/abc123	"Link not found" page	🔴
E23	BC sees only assigned batches	BC	Batches list	Only QA batches assigned to this BC visible	🔴
E24	AO views all sessions across platform	AO	AO → Sessions	All 5 QA sessions visible in one list	🔴
E25	Health check endpoint	Dev	GET /api/v1/health	Returns 200 with server status	🔴

Day 2 — 30 April 2026

Theme: LIVEKIT ADVANCED — All Batch Sizes, All Devices, Connectivity & Resilience

> Day 1 covered every LiveKit feature once in a controlled 1:3 setup on Desktop Chrome. Day 2 stress-tests everything across ALL batch sizes under real load, ALL devices students use, connectivity failure scenarios, and advanced configurations. By end of Day 2, every LiveKit feature has been verified across every environment.

PART 1 — All Batch Sizes Under Load

#	Scenario	Role / Batch	Steps	Expected	Phase
1	1:15 — all 15 students join simultaneously	15 students + Teacher	All 15 open join link at same time → enter room	All 15 tiles render correctly; grid layout adapts; no crash; no missing participant	🔴
2	1:15 — teacher audio stable with 15 connected	Teacher + 15	Run class for 5 min, teacher talks	No audio glitches; all 15 hear teacher clearly throughout	🔴
3	1:15 — teacher screen share to all 15	Teacher	Screen share → select window	All 15 students see the shared screen simultaneously, no black tiles	🔴
4	1:15 — mute all students at once	Teacher	Teacher controls → Mute All	All 15 muted simultaneously; all mic icons update on their tiles	🟡
5	1:30 — all 30 students join	30 students + Teacher	All 30 open links → enter room	All 30 tiles visible; grid scrollable; layout stable; no crash	🔴
6	1:30 — teacher whiteboard visible to 30	Teacher	Open whiteboard → draw circles	All 30 students see drawing in real time; no lag	🔴
7	1:30 — raise hand from 10 students	10 students	All 10 click Raise Hand	Teacher sees all 10 raise-hand notifications; queue shown	🟡
8	1:30 — end class for 30 gracefully	Teacher	End Session → confirm	All 30 disconnected; session status = ended; no hanging connections	🔴
9	Lecture — 50 students join	50 students + Teacher	50 open links → enter	All 50 tiles render in lecture grid; no crash	🔴
10	Lecture — no student can publish audio/video	50 students	Any student tries mic or cam button	All mic/cam controls disabled or hidden for every student	🔴

PART 2 — Connectivity & Resilience

#	Scenario	Role	Steps	Expected	Phase
11	Student drops mid-class → rejoins	Student	Close tab → reopen join link 2 min later	Re-enters room; attendance time accumulates (20 min + 2 min gap + rejoined time)	🔴
12	3 students drop and rejoin simultaneously	3 students	All 3 close tabs at same second → all reopen	All 3 rejoin; no race condition; all 3 attendance records correct	🔴
13	Teacher internet drops briefly	Teacher	Throttle teacher's connection → restore after 20s	Students see "teacher disconnected" placeholder; teacher reconnects; class resumes normally	🔴
14	Long class — 90 min, no timeout	Teacher + 3 students	Run 1:3 class for 90 min continuously	No auto-disconnect; no session timeout; all controls still work at 90-min mark	🔴
15	Student joins on two devices simultaneously	Same student	Phone + laptop both open same join link	Both join OR system handles gracefully (shows "already in session" on 2nd device)	🟡

PART 3 — All Devices & Browsers

#	Scenario	Device / Browser	Steps	Expected	Phase
16	Pre-join lobby — Android Chrome	Android Chrome	Open join link → lobby screen	Lobby loads; mic test works; camera preview shows; Enter button accessible	🔴
17	Pre-join lobby — iOS Safari	iOS Safari	Open join link → lobby	Lobby loads; permission prompts shown correctly; keyboard doesn't overlap Enter button	🔴
18	Full join + mic/cam — Firefox Desktop	Firefox	Open join link → enter room → toggle mic + cam	Room loads; mic/cam toggle works; chat works	🔴
19	Full join + screen share — Safari Mac	Safari Desktop	Join → toggle mic/cam → start screen share	Room loads; controls work; screen share requires permission prompt	🟡
20	Full join — Edge Desktop	Edge	Open join link → enter room	Room loads; all basic controls functional	🔴
21	IE — graceful unsupported page	IE Desktop	Open join link in Internet Explorer	"Browser not supported" page with helpful upgrade message — no crash, no blank page	🔴

#	Scenario	Device / Browser	Steps	Expected	Phase
22	Samsung Internet — join or fallback	Samsung browser	Open join link on Samsung phone	Room loads OR graceful "please use Chrome" message — no crash	🟡
23	Mobile — landscape orientation	Android + iOS	Rotate phone to landscape while in room	UI fills landscape correctly; controls accessible; no layout break	🟡
24	Mobile — overlay auto-hide	Android Chrome	Be in room for 3+ sec without touching	Video controls fade after inactivity	🟡
25	Mobile — overlay reappear on tap	Android Chrome	Tap screen after controls fade	Controls reappear immediately	🟡

PART 4 — AI Monitoring + Advanced Observer Features

#	Scenario	Role / Device	Steps	Expected	Phase
26	AI monitoring — Firefox Desktop	Student on Firefox	Student joins → monitoring initialises	MediaPipe loads silently; no errors in browser console	🔴
27	AI monitoring — iOS Safari	Student on iPhone	Student joins → monitoring initialises	AI monitoring active on mobile Safari; no crashes	🔴
28	AI monitoring — attention alert fires	Student	Student looks away from camera for 30+ seconds	Alert fires in teacher's Monitoring sidebar tab; alert type logged	🟡
29	AI monitoring — writing mode suppresses alerts	Student	Writing mode ON → student looks down to write	NOT flagged as "looking away" — writing-mode exception works	🟡
30	BC records session from CoordinatorLiveView	BC	CoordinatorLiveView → live room → Start Recording	YouTube egress recording starts; REC indicator shows; recording URL available after end	🟡

Day 3 — 1 May 2026

Theme: ENROLLMENT LINK + PAYMENTS + CREDITS — Full Self-Enrollment & Billing Lifecycle

> Day 1 covered manual enrollment. Day 3 tests the self-enrollment path and the complete billing lifecycle. Flow: AO creates link → student fills form → Razorpay payment → account created → AO

> assigns to batch → credits → session gate. These two flows are tested together because enrollment triggers payment, and payment unlocks class access.

PART 1 — Enrollment Link (Tests 1–15)

#	Scenario	Role	Steps	Expected	Phase
1	AO creates enrollment link	AO	Enrollment → Create Link → grade=10, board=CBSE, region=India	Unique `/enroll/[linkId]` URL generated; visible in enrollment links list	🔴
2	AO sends link via WhatsApp button	AO	Click "Send via WhatsApp" → enter student phone	WhatsApp message with link delivered to student phone	🔴
3	Student opens link — Desktop Chrome	Student	Open `/enroll/[linkId]` in Chrome	Form loads; grade/board pre-filled from link config	🔴
4	Student opens link — Android Chrome	Student	Open link on Android phone	Form loads on mobile; all fields accessible; scrollable	🔴
5	Student opens link — iOS Safari	Student	Open on iPhone Safari	Form loads; keyboard does not cover Submit button	🔴
6	Student fills enrollment form	Student	Full Name, Email, Phone, Grade, Board, Region → Next	All fields validated; email format check; required fields enforced	🔴
7	Student proceeds to Razorpay	Student	Click Pay/Next	Razorpay opens with correct amount (grade + board + region fee)	🔴
8	Student completes Razorpay test payment	Student	Select test card → complete	Payment succeeds; account created; success screen shown	🔴
9	Welcome email with credentials sent	Student	Check inbox after payment	Email: login URL, email address, password — arrives within 2 min	🔴
10	Student logs in with received credentials	Student	Login page → use email credentials	Student dashboard loads; sees batch/session area	🔴
11	AO assigns self-enrolled student to batch	AO	AO → Students → pick student → Assign to Batch → 1:3 QA	Student added; join tokens generated; session links emailed	🔴

#	Scenario	Role	Steps	Expected	Phase
12	Student enters session via join link	Student	Open join link → lobby → enter room	Enters room successfully; attendance records start	🔴
13	Enrollment link expired	Student	Open link after 7+ days	"Link expired" page — not a 500 error, not a blank crash	🟡
14	Enrollment link already used	Student	Re-open same used link	"Already enrolled" message	🟡
15	Razorpay payment failure	Student	Use test-failure card	Failure handled gracefully; retry shown; no account created	🔴

PART 2 — Payments, Invoices & Credits (Tests 16–30)

#	Scenario	Role	Steps	Expected	Phase
16	AO generates session-based invoice	AO	Student → Generate Invoice → 1:3 batch, 4 sessions	INV-YYYYMM-##### format; correct amount for batch type × region × board	🔴
17	Invoice email with payment link sent	Student	After generation	Invoice email arrives; contains `/pay/[token]` link	🔴
18	Public payment page — no login required	Student	Open `/pay/[token]` in incognito	Razorpay checkout opens without login; no redirect to login page	🔴
19	Student pays via Razorpay test	Student	Test card → complete	Payment verified; receipt created; credits added	🔴
20	Credits added after payment — correct count	Auto	AO checks student credit balance	Credits match amount paid / fee structure	🔴
21	Receipt email — student + parent	Auto	After payment	Both receive receipt: amount, date, invoice number, receipt number	🔴
22	`/pay/invalid` — graceful error	Student	Open `/pay/invalid123`	"Invalid or expired link" page — not a 500 crash	🔴
23	Invoice already paid — cannot double-pay	Student	Re-open payment link for paid invoice	"Already paid" message; student not charged twice	🟡

#	Scenario	Role	Steps	Expected	Phase
24	Session gate — student WITH credits	Student	Join session link (credits > 0)	Enters room; credit deducted	🔴
25	Session gate — student ZERO credits	Student	Credits = 0 → open join link	Blocked: payment prompt with amount due — cannot enter	🔴
26	Pay from gate prompt → rejoin	Student	Tap "Pay Now" → Razorpay → complete → reopen link	Credits added; student enters room	🔴
27	Skip payment gate flag (AO sets)	AO + Student	AO → `skip_payment_gate = true` → student with 0 credits joins	Enters without payment prompt	🟡
28	Monthly invoice — lecture batch	AO	AO → Invoices → Generate Monthly → lecture batch	All enrolled students receive monthly invoices; correct amounts	🟡
29	Student views own fee history	Student	Student portal → Fees tab	Invoice list, paid/unpaid status, credit balance all correct	🟡
30	AO credit check preview before invoicing	AO	AO → Credit Preview → select batch	Each student's credit balance shown before invoices are generated	🟡

Day 4 — 2 May 2026

Theme: ACADEMIC CALENDAR & SESSION SCHEDULING

> AO sets up the academic calendar, maps topics, and schedules a full week of sessions. BC session scheduling is not yet implemented — AO is the scheduler in the current codebase.

#	Scenario	Role	Steps	Expected	Phase
1	AO/Teacher creates academic calendar	AO	Academic Calendar → New → batch, academic year, start/end dates	Calendar created for the batch	🔴
2	Add curriculum topics to calendar	Teacher/AO	Calendar → Add Topic → subject, chapter, topic name, session number	Topics listed in calendar for the term	🔴
3	View next suggested topic	BC/Teacher	Calendar → Next Topics	System suggests next unfinished topic for each subject	🟡
4	AO schedules session linked to	AO	New Session → pick topic from calendar	Session created, topic linked, visible in	🔴

#	Scenario	Role	Steps	Expected	Phase
	topic			calendar view	
5	AO schedules a full week — Monday	AO	New Session → 1:3 batch, Monday date, Math, Teacher A, 10am, 60 min	Monday session created	🔴
6	AO schedules — Tuesday	AO	Same batch, Tuesday, Science, Teacher B, 11am	Tuesday session created	🔴
7	AO schedules — Wednesday	AO	Same, Wednesday, English	Wednesday created	🔴
8	AO schedules — Thursday	AO	Same, Thursday, Math	Thursday created	🔴
9	AO schedules — Friday	AO	Same, Friday, Science	Friday created	🔴
10	All 5 sessions visible in calendar	AO	AO → Calendar → week view	Mon–Fri sessions shown on correct dates	🔴
11	Students get join links for all 5 sessions	Students	After scheduling	Each student receives 5 join tokens/emails	🔴
12	AO sets up recurring sessions — 4 weeks	AO	Auto Schedule → batch, day=Wednesday, time=10am, duration=60, weeks=4	4 sessions on correct Wednesday dates	🔴
13	Recurring sessions on correct dates	AO	Calendar view	Each recurring session falls on the right day	🔴
14	AO assigns teacher to each session	AO	Session → assign teacher	Each session has a teacher	🔴
15	Multiple teachers per subject (same batch)	AO	Assign Teacher A and Teacher B to Math	AO can assign either teacher per session	🟡
16	AO assigns different teacher to session 2	AO	Session 2 → assign Teacher B	Teacher B sees session 2, Teacher A does not	🟡
17	AO reschedules one session	AO	Session → Edit → new date/time	Updated, students notified	🔴
18	AO cancels one session	AO	Session → Cancel → reason	Cancelled, students + parents notified	🔴

#	Scenario	Role	Steps	Expected	Phase
19	Cancel one session in recurring series	AO	Cancel week 2 only	Weeks 1, 3, 4 unaffected	🟡
20	Session without teacher — warning	AO	Create session without assigning teacher	Warning: "teacher not assigned"	🟡
21	Past date rejection	AO	Try to schedule for yesterday	Error: cannot schedule in the past	🟡
22	Teacher daily limit (4 sessions)	AO	Try to schedule 5th session for same teacher same day	Error: daily limit reached	🟡
23	Student gets all week's join links	Student	Check email	5 separate join links received, one per session	🔴
24	New student added mid-week	AO	AO adds student to batch after Mon–Fri scheduled	Student gets backfilled tokens for Tue–Fri (remaining sessions)	🔴
25	New student's backfilled links work	New Student	Student opens backfilled join link	Enters room successfully	🔴
26	Add topic from calendar to existing session	Teacher/AO	Session → link topic from academic calendar	Topic shown in session details	🟡
27	Topic marked taught after session	Auto/Teacher	After session ends	Topic marked as covered in academic calendar	🟡
28	Next topic suggestion updates	Teacher/AO	View next topics after session completed	System now suggests next topic in sequence	🟡
29	BC calendar monthly view	BC	Calendar → Monthly	All sessions shown on correct dates, colour by status	🟢
30	AO views sessions across all batches	AO	AO dashboard → sessions	All batches' sessions visible to AO	🟡

AI EXAM GENERATION PIPELINE (Groq Cloud)

> AO uploads PDF/image files per topic → triggers AI generation → reviews questions → teacher uses them in session exams.

#	Scenario	Role	Steps	Expected	Phase
31	AO creates exam topic	AO	AO → Exam Questions → New Topic → subject=Math, grade=Class 10, board=CBSE, chapter=Algebra, category=topic	Topic created, status=draft; appears in list	🔴
32	AO uploads files to topic	AO	Topic → Upload Files → attach PDF (≤20 MB) + PNG image	Files uploaded; list shows name, size, type	🔴
33	AO triggers AI generation	AO	Topic → Generate Questions → page_range="1-5" → confirm	Progress indicator shown; Groq API called server-side	🔴
34	AI returns questions — correct format	Auto	After generation completes	Questions list: text, 4 options, correct_answer index, marks, difficulty all populated	🔴
35	AO reviews and approves questions	AO	Review each question → Approve → status = ready	Topic status changes to "ready"	🔴
36	Teacher selects ready topic in session exam	Teacher	In-class Exam tab → Select Topic	Only "ready" topics visible; teacher selects one	🔴
37	Teacher pushes AI-generated exam to students	Teacher	Exam → Push to Students	Exam overlay appears for all students in room	🔴

QUESTION BANK (Manual)

> Teacher/AO manually create reusable MCQ questions tagged by subject/grade/difficulty. Used across multiple session exams.

#	Scenario	Role	Steps	Expected	Phase
38	Teacher creates question in question bank	Teacher	Teacher → Questions tab → New Question → subject, grade, text, 4 options, correct_answer, difficulty	Question saved and tagged; appears in bank	🔴
39	AO creates question	AO	AO → Exam Questions → New Question	Saved; visible to teachers of same subject/grade	🟡
40	Filter questions by subject, grade, difficulty	Teacher	Questions list → apply filters	List narrows to matching questions only	🟡

#	Scenario	Role	Steps	Expected	Phase
41	Teacher picks questions from bank for session exam	Teacher	In-class Exam → Pick from Question Bank → select 5	Selected questions added to exam	🔴
42	Student takes exam built from question bank	Student	Exam overlay appears	Questions display correctly; student can answer and submit	🔴

TEACHING MATERIALS

> AO/Teacher uploads files (PDFs, videos, slides, archives) → assigns to one or many batches → students download.

#	Scenario	Role	Steps	Expected	Phase
43	AO uploads PDF material to a batch	AO	AO → Materials tab → Upload → Title, PDF file, assign to 1:3 batch	Material saved; visible in list with name, size, batch	🔴
44	AO assigns same material to multiple batches	AO	Material → Assign to Batch → add 1:15 batch	Material linked to both batches	🟡
45	Student sees materials for their batch	Student	Student → Classes or Materials section	Materials for student's batch listed with download button	🔴
46	Student downloads material	Student	Click material → download	File downloads; correct filename and format	🔴
47	Teacher uploads material from teacher dashboard	Teacher	Teacher → Materials tab → Upload	Material uploaded; linked to teacher's assigned batches	🟡
48	AO removes material from batch	AO	Material → Unassign from batch	Material no longer visible to that batch's students	🟡

STUDENT AVAILABILITY

> Students submit preferred time slots → AO/Coordinator views for optimal scheduling.

#	Scenario	Role	Steps	Expected	Phase
49	Student submits availability slot	Student	Student → Requests → Availability → Day=Monday, Start=10:00, End=12:00, preference=preferred	Slot saved; visible in student's profile	🟡

#	Scenario	Role	Steps	Expected	Phase
50	Student adds multiple slots	Student	Repeat for Tuesday and Wednesday	All 3 slots stored correctly	🟡
51	AO views student availability	AO	AO → Students → click student → Availability tab	All submitted slots shown with day/time/preference	🟡
52	AO uses availability to inform scheduling	AO	Cross-reference availability while creating sessions	AO can see which days students prefer before picking session time	🟢

Day 5 — 3 May 2026

Theme: PRE-CLASS FLOW — Reminders, Lobby, Teacher Enters, BC Goes Live

> From "15 minutes before class" to "teacher is live and ready." This is the daily handshake.

#	Scenario	Role	Steps	Expected	Phase
1	Session reminder — WhatsApp (15 min before)	Auto	Schedule session 20 min from now, wait 5 min	WhatsApp arrives exactly at 15 min mark for student	🔴
2	Session reminder — email (15 min before)	Auto	Same session	Email reminder also received	🔴
3	Parent receives reminder	Auto	Same session	Parent WA/email received too	🟡
4	Student opens join link from reminder	Student	Click link in WA message	Pre-join lobby loads	🔴
5	Pre-join lobby loads	Student	Lobby screen	Device check screen shown before room entry	🔴
6	Mic test in lobby	Student	Speaks in lobby	Audio meter moves	🔴
7	Camera test in lobby	Student	Enables camera	Video preview shown	🔴

#	Scenario	Role	Steps	Expected	Phase
8	Mic denied by browser — lobby handles it	Student	Browser popup → deny mic	Lobby shows "mic denied" warning, can still enter	🔴
9	Camera not found — lobby handles it	Student	Device has no camera	Lobby shows "no camera found", can still enter	🔴
10	Poor network warning in lobby	Student	Throttle connection	Lobby shows "poor connection" warning	🟡
11	Student skips device check	Student	Click "Enter anyway"	Enters room without completing checks	🟡
12	Student enters room — waits for teacher	Student	Enter before teacher goes live	Waiting state shown, not an error	🟡
13	Teacher login	Teacher	Login on Chrome	Dashboard, today's session card visible	🔴
14	Teacher pre-join lobby	Teacher	Click Join → lobby	Device check for teacher	🔴
15	Teacher enters room	Teacher	Enter from lobby	Room loads, teacher's own video tile shows	🔴
16	BC is online (heartbeat active)	BC	BC logs in, stays on dashboard	BC heartbeat active, coordinator-status = online	🔴
17	Teacher sends go-live request	Teacher	Click Go Live	Request sent to BC	🔴
18	BC sees go-live approval dialog	BC	BC dashboard / notification	Dialog appears with approve/deny	🔴
19	BC approves go-live	BC	Click Approve	Room status → live	🔴

#	Scenario	Role	Steps	Expected	Phase
20	Students auto-enter after go-live	Students	Were waiting in room	Students who were waiting now see the live room	🟡
21	BC offline — go-live blocked	BC	BC logs out → teacher → Go Live	Request either queued or teacher sees "coordinator offline"	🔴
22	Go-live skip toggle (bypass approval)	AO	Set `go_live_skip_coordinator = true` → teacher → Go Live	Goes live immediately, no BC needed	🔴
23	Early start toggle	AO	Set `allow_go_live_before_schedule = true` → teacher goes live 10 min early	Room goes live before scheduled time	🟡
24	BC denies go-live request	BC	BC clicks Deny	Teacher sees denied, must wait or re-request	🟡
25	Teacher denied — re-requests	Teacher	Teacher re-sends go-live request	BC gets second notification	🟡
26	Student joins before class (no teacher yet)	Student	Opens link 10 min before scheduled time	Lobby or waiting room shown, not error	🟡
27	Student joins via email link — no login needed	Student	Click pre-auth token link	Enters directly without login screen	🔴
28	Join link from reminder works on Android	Student	Open WA → tap link on Android	Pre-join lobby opens in mobile browser	🔴
29	Join link from reminder works on iOS	Student	Open WA → tap link on iPhone	Pre-join lobby opens in Safari	🔴
30	Room shows "live" to BC	BC	BC checks live dashboard	Room card shows LIVE	🔴

#	Scenario	Role	Steps	Expected	Phase
	after approval			status	

Day 6 — 5 May 2026

Theme: LIVE CLASS — Core Features (1:1 and 1:3, Desktop)

> Class is live. Test every feature a teacher and student use in a normal class.

#	Scenario	Steps	Expected	Phase
1	Teacher mic on/off	Teacher toggles mic	Icon updates, student hears change instantly	🔴
2	Teacher camera on/off	Teacher toggles camera	Video tile shows/hides, student sees change	🔴
3	Student mic on/off (1:1 and 1:3)	Student toggles mic	Teacher hears change	🔴
4	Student camera on/off	Student toggles camera	Teacher sees change	🔴
5	Both teacher and student video visible	Both cameras on	Each sees the other clearly	🔴
6	Teacher screen share — Chrome	Screen share → select entire screen	Student sees teacher's full screen	🔴
7	Teacher screen share — specific window	Teacher shares one app window	Only that window shown	🟡
8	Stop screen share	Teacher stops	Returns to camera view, no black screen	🔴
9	Teacher whiteboard	Opens whiteboard → draws	Student sees drawing in real time	🟡
10	Switch from screen share to whiteboard	Teacher switches	Smooth transition	🟡
11	Student raises hand	Student clicks ✋	Teacher sees notification + sound	🔴
12	Teacher lowers hand	Teacher dismisses	Notification cleared	🟡
13	Multiple raises (1:3)	All 3 students raise hand	All 3 in queue on teacher side	🟡
14	Teacher mutes student	Right-click tile → Mute	Student muted, icon updates both sides	🔴
15	Teacher unmutes student	Right-click → Unmute	Student mic activates	🟡
16	Chat — student sends	Student types + send	Seen by teacher and all students	🔴

#	Scenario	Steps	Expected	Phase
17	Chat — teacher sends	Teacher sends	All students see it	🔴
18	Contact violation — phone in chat	Student types phone number	Blocked / warning shown	🟡
19	Teacher spotlight student	Teacher clicks student → spotlight	Tile enlarges for teacher view	🟡
20	Teacher removes student	Teacher removes student from room	Student disconnects, sees "removed" message	🔴
21	Removed student tries to rejoin	Student opens link again	"Removed from session" — blocked	🔴
22	Teacher announcement	Teacher → Announcement bar → types	Students see/hear announcement overlay	🟡
23	Time warning dialog	Approach scheduled end time	Warning dialog appears for teacher	🟡
24	Teacher creates homework in room	Homework tab → create assignment	Homework visible to students	🟡
25	Teacher pushes session exam	Exam tab → start exam	Exam pushed to all students in room	🔴
26	Student answers session exam	Student selects answers	Answers saved, 60s timer shown	🔴
27	Exam auto-graded	All submit	Scores calculated, grade letters assigned	🔴
28	Teacher ends class	End Class → confirm	All disconnected, status → ended	🔴
29	Post-class feedback dialog	Session ends	Feedback/rating dialog auto-appears for student	🟡
30	Attendance recorded	After session	All students marked with correct status and duration	🔴

Day 7 — 7 May 2026

Theme: LIVE CLASS — All Batch Types & All Devices

> Same class, scaled up to all batch sizes and all devices students actually use.

#	Scenario	Batch / Device	Expected	Phase
1	1:15 — all 15 students join	1:15	All 15 visible, grid adapts, no crash	🔴
2	1:15 — teacher audio reaches all	1:15	Stable audio to 15 students	🔴

#	Scenario	Batch / Device	Expected	Phase
3	1:15 — teacher screen share	1:15	All 15 see the share	🔴
4	1:15 — mute all / unmute one	1:15	Mute all works; unmute one only activates that student	🟡
5	1:30 — all 30 students join	1:30	All 30 visible, grid scrollable, no crash	🔴
6	1:30 — teacher audio stable	1:30	Stable to 30 students	🔴
7	1:30 — end class gracefully	1:30	All 30 disconnected, session ended	🔴
8	Lecture — student mic/cam hidden	Lecture	No mic/camera controls for students	🔴
9	Lecture — 50 students join	Lecture	All 50 enter, no crash	🔴
10	Lecture — student raises hand	Lecture	Raise hand works even with mic disabled	🔴
11	Lecture — media request (student requests mic)	Lecture	Teacher sees request → approve/deny	🟡
12	Student joins — Desktop Chrome	Desktop Chrome	Room loads, video/audio works	🔴
13	Student joins — Desktop Firefox	Desktop Firefox	Same	🔴
14	Student joins — Desktop Edge	Desktop Edge	Same	🔴
15	Student joins — Desktop Safari (Mac)	Desktop Safari	Room loads, permissions work	🔴
16	Student joins — Internet Explorer	IE	Graceful "browser not supported" page — no crash	🔴
17	Student joins — Android Chrome	Android Chrome	Room loads, video/audio works	🔴
18	Student joins — iOS Safari	iOS Safari	Room loads, mic/cam permissions correct	🔴
19	Student joins — Samsung Internet	Samsung Internet	Loads or graceful fallback	🟡
20	Student via email link (no login)	Any device	Pre-auth token works, enters without login	🔴
21	Overlay auto-hide (mobile)	Android/iOS	Controls fade after 3s inactivity	🟡
22	Tap to reveal overlays	Android/iOS	Controls reappear on tap	🟡
23	Landscape rotation	Android/iOS	UI fills landscape, no layout breaks	🟡
24	Teacher mic/cam — iOS	iOS Safari	Toggles correctly	🔴

#	Scenario	Batch / Device	Expected	Phase
25	Teacher mic/cam — Android	Android Chrome	Toggles correctly	🔴
26	Screen share — Firefox	Desktop Firefox	Works correctly	🔴
27	Screen share — Safari	Desktop Safari	Works with permissions	🟡
28	AI monitoring loads — Chrome	Chrome Desktop	Loads silently	🔴
29	AI monitoring loads — iOS Safari	iPhone	Loads on mobile Safari	🔴
30	AI monitoring loads — Android	Android Chrome	Loads on Android	🔴

Day 8 — 8 May 2026

Theme: LIVE CLASS — Edge Cases, Teacher Controls & Advanced Features

> Things that happen unexpectedly. Advanced teacher tools.

#	Scenario	Steps	Expected	Phase
1	Student drops mid-class (tab closed)	Student closes tab → reopens link 2 min later	Rejoins seamlessly, attendance time accumulates	🔴
2	Teacher internet drops briefly	Simulate brief disconnect	Students see "teacher disconnected" → teacher reconnects	🔴
3	Teacher mic fails mid-class	Mic drops unexpectedly	Students see mic-off icon; teacher can re-enable	🔴
4	Teacher camera drops mid-class	Cam disconnects	Students see avatar tile; teacher can re-enable	🔴
5	3 students join simultaneously	All 3 click join at same second	All 3 enter, no race condition, all attendance recorded	🔴
6	Student joins on two devices	Same student: phone + laptop	Second join shown or blocked gracefully	🟡
7	Long class — 90 minutes	Run 1:1 class for 90 min	No timeout, no auto-disconnect	🔴
8	Session extended beyond scheduled time	Teacher continues past end	Timer shows overtime, session stays live	🟡

#	Scenario	Steps	Expected	Phase
9	Session extension request	Student requests more time → teacher approves	Session extended	🟡
10	Selective session extension	Extend for student A, end for student B	A stays, B disconnects	🟡
11	Teacher requests early end — BC approves	Teacher → End Early → BC approves	Class ends	🟡
12	Teacher requests early end — BC denies	BC denies	Class continues	🟡
13	Virtual background — blur	Teacher → background blur	Blur applied, smooth video	🟢
14	Virtual background — image	Teacher sets background image	Image shown behind teacher	🟢
15	Video quality selector	Teacher changes quality to Low	Bitrate drops	🟢
16	Recording start	Teacher → start recording	Recording begins, REC indicator shown	🟡
17	Recording stop	Teacher stops	URL saved	🟡
18	Teacher tablet (screen-share role)	Teacher joins second device as teacher_screen	Tablet screen visible in room	🟡
19	Student posts doubt	Doubt panel → type question → submit	Doubt logged, teacher notified	🟡
20	Teacher responds to doubt	Teacher → Doubts → responds	Student sees reply	🟡
21	Student submits homework	Homework panel → upload file → submit	Submission received	🟡
22	Teacher views all homework submissions	Teacher → Homework	All submissions listed	🟡
23	Report teacher button	Student → report teacher → fills form → submit	Report saved, AO notified	🟡
24	Teacher controls toggle — disable exam	AO sets `allow_exam_push = false` → teacher UI	Exam button hidden for teacher	🟡
25	Teacher controls toggle — disable recording	AO sets `allow_recording = false`	Recording button hidden	🟡
26	Teacher controls toggle — disable homework	AO sets `allow_homework_create = false`	Homework button hidden	🟡
27	Contact violation — Telegram handle	Student types @handle in chat	Blocked/warned	🟡

#	Scenario	Steps	Expected	Phase
28	Contact violation — Instagram link	Student types Instagram URL	Blocked/warned	🟡
29	Session ends — all students disconnected gracefully	Teacher ends with students in room	No hanging connections	🔴
30	Session status after end	BC/AO checks	Status = ended, correct start/end times	🔴

Day 9 — 9 May 2026

Theme: POST-CLASS — Attendance, Exam Results, Feedback, Reports

> Everything that happens after the class ends.

#	Scenario	Steps	Expected	Phase
1	Present — on-time join	Student joins within 5 min	Status = present	🔴
2	Late — joins 10 min after start	Student joins at 10 min	Status = late, late_by_sec = 600	🔴
3	Absent — never joined	Student never opens link	Status = absent after session ends	🔴
4	Left early — short session	Student joins 10 min, leaves, no rejoin	Duration = 10 min, status = left_early or present	🔴
5	Duration accumulates across rejoins	Join 20 min → leave → rejoin 15 min	Total ~35 min	🔴
6	Attendance for 30 students	30 students join at various times	All 30 entries, none missing	🔴
7	Teacher manual attendance override	Teacher → Attendance tab → mark "excused"	Override saved	🟡
8	BC views attendance post-session	BC → Session → Attendance	All students listed with duration, status, join time	🟡
9	AO views attendance	AO → Batch → Session → Attendance	Same data visible to AO	🟡
10	Parent views child's attendance	Parent → Attendance	Correct status for this session	🟡

#	Scenario	Steps	Expected	Phase
11	Student views own attendance	Student → Attendance	Correct status shown	🟡
12	Exam result email — student	After auto-grade	Student receives result email with score + grade letter	🔴
13	Exam result email — parent	Same	Parent also receives result email	🔴
14	Teacher views all exam scores	Teacher → Exam Results tab	All student scores with grade letters	🟡
15	Session exam — tab switch violation	Student switches tab during exam	Violation logged, teacher notified	🟡
16	Session exam — auto-submit on timeout	No answer → timer runs out	Auto-submitted with current state	🟡
17	Post-session feedback saved	Student submits rating	Rating stored, visible to AO/owner	🟡
18	No daily AI monitoring report to parent	Session ends with monitoring events	Parent does NOT get daily email	🔴
19	Monthly report only	Check report config	Only monthly frequency configured	🔴
20	Monthly monitoring report content	View generated monthly report	Attention score, alert breakdown, time-in-states shown	🟢
21	Session report generated	AO/BC → Session → Report	Report shows summary: attendance, exam scores, duration	🟡
22	Batch report generated	BC → Batch Report	Attendance trends across sessions shown	🟡
23	Attendance after teacher ends early	BC approves early end	Attendance closed at actual end time, not scheduled	🟡
24	Attendance export PDF	BC → Export	PDF downloads correctly	🟢
25	Topic marked taught after session	Academic Calendar	Session's linked topic marked as covered	🟡
26	Next topic suggestion updated	Teacher/AO	Next Topics view	Shows next in sequence, not the just-covered one

#	Scenario	Steps	Expected	Phase
27	Student session history updated	Student	Student → My Sessions	Just-completed session appears with status
28	Teacher session history	Teacher	Teacher → My Sessions	Session shown with attendance count, exam results
29	Recurring session: attendance across 4 weeks	Run 4 sessions	History shows all 4 sessions per student	🟡
30	AO full reports — platform-wide	AO	AO → Reports	Attendance, revenue, teacher perf, student progress all available

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STANDALONE EXAMS (Assigned to Batch, Not Live-Pushed)

> Teacher creates exam in dashboard → assigns to batch with due date → students take it from their Exams tab.

#	Scenario	Role	Steps	Expected	Phase
31	Teacher creates standalone exam	Teacher	Teacher → Exams tab → New Exam → title, add questions from question bank	Exam created, status=draft	🔴
32	Teacher assigns exam to batch with due date	Teacher	Exam → Assign to Batch → 1:15 QA batch → due in 3 days	All students in batch can see the exam	🔴
33	Student sees pending exam in dashboard	Student	Student → Exams tab	Exam listed with title, due date, status=pending	🔴
34	Student takes standalone exam	Student	Student → Exams → Start → answer MCQs → Submit	Auto-graded; score and grade letter shown immediately	🔴
35	Teacher reviews all submissions	Teacher	Teacher → Exams → click exam → Submissions	All student attempts listed: score, grade, time taken	🟡
36	AO views all exams platform-wide	AO	AO → Exam Questions / Exams list	All exams: creator, question count, attempt count visible	🟡

TEACHER RATINGS & FEEDBACK

> Students rate the teacher after each session via the post-class feedback dialog.

#	Scenario	Role	Steps	Expected	Phase
37	Student submits post-session rating	Student	Post-class feedback dialog → star rating (1–5) + optional comment → Submit	Rating stored; associated with teacher + session	🔴
38	Teacher sees cumulative ratings in Ratings tab	Teacher	Teacher → Ratings tab	Average stars, per-session breakdown, comments all listed	🟡
39	AO sees teacher rating summary	AO	AO → Teachers → click teacher profile	Rating history with averages visible	🟡
40	Owner sees all teacher ratings	Owner	Owner → Teacher Reports	All teachers listed with average rating and trend	🟢

Day 10 — 12 May 2026

Theme: STUDENT MANAGEMENT — Add Mid-Batch, Block, Remove, Ongoing Operations

> Ongoing daily operations: students join mid-series, get blocked, removed, re-added.

#	Scenario	Steps	Expected	Phase
1	Add student mid-recurring batch	AO adds student after 4-week recurring series exists	Student gets backfilled tokens for all remaining future sessions	🔴
2	New student joins backfilled session	New student opens backfilled join link	Enters room successfully, attendance recorded	🔴
3	Add 3rd student to 1:3 batch mid-series	After recurring sessions exist → add student 3	Student 3 gets all future tokens	🔴
4	New student's gate check	New student with credits → joins	Enters; without credits → blocked	🔴
5	Block student	AO → Student → Block	Account suspended	🔴
6	Blocked student login	Blocked student tries to login	"Account suspended" — not blank	🔴
7	Blocked student join link	Blocked student tries join link	"Account suspended" gate	🔴

#	Scenario	Steps	Expected	Phase
8	Unblock student	AO → Unblock	Login and join restored	🔴
9	Remove student from batch	AO → Batch → Remove student	Student removed, future tokens invalidated	🔴
10	Removed student's old join link	After removal → old link	"No longer enrolled in this batch" — not a crash	🔴
11	Remove student mid-recurring batch	5 sessions remain → remove student	All 5 future tokens invalidated	🔴
12	Re-add removed student	AO re-adds same student	Back in batch, tokens backfilled again	🟡
13	Delete student from system	AO deletes student entirely	All join links invalid, account gone	🔴
14	Teacher deactivated	AO → Users → Teacher → deactivate	Teacher cannot login	🔴
15	Deactivated teacher's sessions flagged	BC checks sessions	Flagged: "teacher unavailable"	🔴
16	BC assigns substitute teacher	BC → flagged session → assign substitute	Substitute sees session	🔴
17	Teacher reactivated	AO → reactivate	Teacher can login and teach	🔴
18	Teacher leave request	Teacher → Apply Leave	Leave request submitted	🔴
19	HR approves teacher leave	HR → Teacher Leave → approve	Approved, sessions flagged	🔴
20	BC assigns substitute after leave approved	BC → flagged session → assign substitute	Substitute assigned	🔴

#	Scenario	Steps	Expected	Phase
21	Student full profile view	AO	AO → Students → click any student	Name, email, grade, board, region, credits, attendance %, batch list all visible
22	Multiple teachers per subject — AO assigns per session	AO	1:15 batch → Sessions → assign Teacher A to Monday, Teacher B to Wednesday	Each teacher sees only their own assigned sessions in dashboard
23	Edit student details	AO → Student → Edit phone/grade	Updated correctly	🟡
24	Add parent email to student	AO → Student → add parent email	Parent linked, receives future emails	🟡
25	Student self-registration blocked	Student goes to /login	No self-registration option	🔴
26	Student profile page	Student login → profile	Name, batch, attendance, credits, exams all shown	🟡
27	Parent files complaint	Parent → Complaints → file	Submitted, AO notified	🟡
28	AO views complaint	AO → dashboard	Complaint visible	🟡
29	CRM verifies students	AO	AO checks students from smartuplearningventures@gmail.com	Students verifiable and assignable
30	Ghost audit log entry	AO/Ghost observes class → leaves	ghost_audit_log entry created	🟡

TEACHER LEAVE — FULL 5-STEP WORKFLOW

> Tests 18–20 above cover the basic path. These cover the complete approval chain with email notifications at every step. Flow: Teacher submits → AO reviews affected sessions → AO sets resolution plan → HR approves → AO confirms.

#	Scenario	Role	Steps	Expected	Phase
31	Teacher submits leave request	Teacher	Teacher → Leave Requests → Apply Leave → start_date, end_date, leave_type=sick, reason	Request created, status=pending_ao; AO notified	🔴
32	AO sees pending leave request	AO	AO → Leave Requests tab	New request listed: teacher name, dates, affected session count	🔴
33	AO views affected sessions	AO	Open request → affected sessions	All sessions within the leave date range listed	🔴
34	AO assigns substitute for each session	AO	Affected session → assign substitute teacher	Substitute field updated; resolution plan saved	🔴
35	AO forwards to HR with resolution plan	AO	Request → Forward to HR → add notes	Status = pending_hr; HR notified by email	🔴
36	HR reviews and approves plan	HR	HR → Leave Requests → open → Approve	Status = approved; AO notified by email	🔴
37	AO confirms approved leave	AO	Request → Confirm	Status = confirmed; substitute teacher receives session notification	🔴
38	Substitute teacher sees new session	Sub Teacher	Login → My Sessions	New session with original batch appears; original teacher name shown for reference	🔴
39	Students notified of teacher change	Students	Check email/WA	Notification: session date, new teacher name, subject	🔴
40	Teacher withdraws pending request	Teacher	Teacher → Leave Requests → Withdraw (before AO confirms)	Status = withdrawn; sessions unaffected; HR notified	🟡

SESSION REQUESTS WORKFLOW

> Students, parents, or teachers submit reschedule/cancel requests → AO approves/rejects → notification chain fires.

#	Scenario	Role	Steps	Expected	Phase
41	Student submits reschedule request	Student	Student → Requests tab → New Request → type=reschedule, session, reason, proposed_date	Request submitted, status=pending; AO notified by email	🔴
42	Parent submits reschedule request	Parent	Parent → Requests tab → New Request	Request submitted on behalf of child	🟡
43	AO views all pending session requests	AO	AO → Leave Requests / Requests section	Pending requests listed: type, requester role, session name, proposed date	🔴
44	AO approves → session rescheduled	AO	Request → Approve → confirm	Session time updated; all enrolled students + parents notified via WA + email	🔴
45	AO rejects → requester notified with reason	AO	Request → Reject → enter rejection_reason	Status = rejected; email with reason sent to requester	🔴
46	Teacher submits cancel request	Teacher	Teacher → via dashboard → cancel session request	Request created; AO reviews; teacher notified of outcome	🟡

ADMISSIONS WORKFLOW

> Structured pipeline: enquiry → registered → fee_confirmed → allocated → active.

#	Scenario	Role	Steps	Expected	Phase
47	AO creates admission request	AO	AO → Admissions → New → name, grade, board, contact	Admission record created, status=enquiry	🟡
48	AO advances to registered	AO	Admission → Update Status → registered	Status updated; timestamp recorded	🟡
49	AO advances to fee_confirmed	AO	Update → fee_confirmed	Status updated	🟡
50	AO allocates to batch	AO	Update → allocated → link batch	Status = allocated; batch linked to record	🟡
51	AO marks as active	AO	Update → active	Student fully onboarded; record status = active	🟡

#	Scenario	Role	Steps	Expected	Phase
52	HR views admission pipeline	HR	HR login → view Admissions	All admission records visible with current status and count per stage	🟡

Day 11 — 13 May 2026

Theme: AI MONITORING, GHOST MODE & BC LIVE MONITORING

> Silent observation, attention tracking, alert policies.

#	Scenario	Steps	Expected	Phase
1	AI monitoring loads — Chrome	Desktop Chrome student	MediaPipe loads silently, no errors	🔴
2	AI monitoring loads — Firefox	Desktop Firefox	No errors	🔴
3	AI monitoring loads — Safari Mac	Desktop Safari	Works or graceful disable	🔴
4	AI monitoring loads — iOS Safari	iPhone	Active on mobile Safari	🔴
5	AI monitoring loads — Android Chrome	Android	Active on Android	🔴
6	Short blink — NO alert	Student blinks normally	No alert	🟡
7	Sustained eyes closed (3+ min) — alert	Student closes eyes 3+ min	Alert in teacher monitoring tab	🟡
8	Looking away — alert	Student looks away continuously	Alert after threshold	🟡
9	Not in frame	Student leaves camera view	"Not in frame" event logged	🟡
10	Tab switch detection	Student switches browser tab	Tab-switch shown to teacher	🟡
11	Writing aware mode ON	Student writes notes (mode enabled)	NOT flagged as "looking away"	🟡
12	Exam strict mode	Exam running + strict mode on	Stricter alerts during exam	🟡
13	Mobile relaxed thresholds	Student on phone + mode enabled	Fewer false alerts on mobile	🟡
14	Low visibility state	Student in poor lighting	"Low visibility" state, not false alert	🟡

#	Scenario	Steps	Expected	Phase
15	Teacher sees monitoring tab	Teacher → Monitoring sidebar tab	Real-time alerts shown	🟡
16	BC sees monitoring alerts	CoordinatorLiveView → Monitoring	Same alerts	🟡
17	Dismiss an alert	Teacher/BC clicks dismiss	Alert removed from list	🟡
18	No daily report to parent	Session ends	Parent does NOT receive daily email	🔴
19	Monthly report confirmed	Check config	Monthly frequency only	🔴
20	Ghost (Owner) joins live session	Owner enters live room	Invisible to all, can see everything	🔴
21	Ghost not in participant list	Teacher/students check list	Ghost not listed, count unaffected	🔴
22	Ghost sees all student tiles	GhostView	All participant tiles shown	🔴
23	Ghost view modes switch	Control → Teacher → Student	Layout changes correctly	🟡
24	Ghost/AO is view-only in classroom	AO ghost → tries to unmute or publish	Audio/video publish blocked — ghost is always view-only	🔴
25	Parent ghost — view only	Parent joins as ghost	Cannot publish or announce	🟡
26	BC live monitoring dashboard	BC → Live	All active rooms for assigned batches listed	🔴
27	BC enters room from dashboard	BC clicks live room card	CoordinatorLiveView loads, BC hidden	🔴
28	BC sees all participants	Inside CoordinatorLiveView	Full grid + attention states visible	🔴
29	BC starts recording	CoordinatorLiveView → Record	YouTube egress starts	🟡
30	Ghost audit log entry	AO observes class → leaves	ghost_audit_log entry created with room, observer, duration	🟡

AI MONITORING PARAMETER TUNING

> Teacher configures alert thresholds per-session from the Monitoring tab inside the classroom.

#	Scenario	Role	Steps	Expected	Phase
31	Teacher opens monitoring settings in classroom	Teacher	Inside classroom → Monitoring tab → Settings icon	Tuning panel opens with threshold controls	🟡
32	Teacher adjusts writing-mode look-down angle threshold	Teacher	Slider for writing-mode angle → increase value	Value saved; applies to this session	🟡
33	Teacher saves tuning config	Teacher	Save Settings	Confirmation shown; config persisted for this teacher	🟡
34	Reduced false alerts after tuning	Teacher + students	Run 15-min class with active note-taking	Writing-mode false alerts suppressed; genuine attention alerts still fire	🟡

GHOST MODE — BY BATCH, BY TEACHER & OVERSIGHT DASHBOARD

#	Scenario	Role	Steps	Expected	Phase
35	Ghost → filter sessions by batch	Ghost/Owner	Ghost → By Batch tab → select batch	All live sessions for that batch listed	🔴
36	Ghost → filter sessions by teacher	Ghost/Owner	Ghost → By Teacher tab → select teacher	Live sessions where that teacher is active shown	🟡
37	Ghost Oversight dashboard (`/ghost/monitor`)	Ghost	Ghost → Oversight nav item	Cross-session monitoring view — all live rooms visible at once	🔴
38	Ghost enters room from By Batch view	Ghost	Click live room card → enter	GhostView loads; ghost invisible to all participants; audit log entry created	🔴

BC TEACHER REPORTS

#	Scenario	Role	Steps	Expected	Phase
39	BC views teacher performance report	BC	BC dashboard → Teacher Reports tab	Each teacher's session count, attendance rate, avg exam score — filtered to assigned batches only	🟡
40	BC exports teacher report	BC	Teacher Reports → Export	CSV or PDF downloads with correct data	🟢
41	AO views teacher reports platform-	AO	AO → Teacher Reports	All teachers; filterable by batch, subject, date range	🟡

#	Scenario	Role	Steps	Expected	Phase
	wide				

Day 12 — 14 May 2026

Theme: OWNER, HR, SALES & PARENT PORTALS

> Every remaining role tested end to end.

#	Scenario	Role	Steps	Expected	Phase
1	Owner login	Owner	Login → dashboard	Full platform: all users, batches, stats	🔴
2	Owner sets custom permission for BC	Owner	Permissions → BC user → add `fees_view = true`	Permission saved	🔴
3	Custom permission applies to that BC	BC	BC logs in → fees section	Fee data now visible	🔴
4	Owner views AO users	Owner	Owner → Admins → view AO user list	All AO users visible, read-only	🟡
5	Owner views global reports	Owner	Owner → Reports	All types: revenue, attendance, teacher performance, student progress — read-only	🔴
6	Owner views system settings	Owner	Owner → System	Config visible, read-only — no edits	🟡
7	HR login	HR	Login → HR dashboard	Users, attendance, payroll visible	🔴
8	HR resets teacher password	HR	HR → Users → Teacher → Reset Password	New password sent, teacher can login	🔴
9	HR approves teacher leave	HR	Teacher Leave → approve	Leave approved, sessions flagged	🔴
10	Teacher salary live tracking	Teacher	Teacher → Salary	Earnings updated after each session	🟡
11	HR generates payslip	HR	Payroll → generate payslip	Payslip created	🟡
12	Payslip PDF download	HR	Download payslip PDF	PDF correct, proper amounts	🟡

#	Scenario	Role	Steps	Expected	Phase
13	Sales login	Sales	Login → Sales dashboard	Leads, pipeline, activities visible	🔴
14	Sales sends demo link	Sales	Sales → Lead → Send Demo Link	WA message sent to prospect	🔴
15	Sales views leads	Sales	Sales → Leads	Full lead list with status	🟡
16	Facebook lead sync	Sales	Sales → FB Sync	Leads from Facebook imported	🟢
17	Parent login	Parent	Login → Parent dashboard	Child's sessions, attendance, fees visible	🔴
18	Parent views child attendance	Parent	Parent → Attendance	Correct status for recent sessions	🔴
19	Parent views child exam results	Parent	Parent → Exams	Exam results shown	🟡
20	Parent views child fees	Parent	Parent → Fees	Invoice history, credit balance	🟡
21	Parent files complaint	Parent	Complaints → file	Complaint submitted, AO notified	🟡
22	Open Classroom — create	AO	Open Classroom → New → subject, grade	OC created, shareable URL generated	🟡
23	Open Classroom — public join	Anyone	Open `/open-classroom/[token]` without login	Joins as viewer	🟡
24	Open Classroom — join approval	Host	OC with approval → request join → host approves	Works	🟡
25	Conference — create	Teacher	Conference → create	Conference room created	🟡
26	Conference — join via token	Anyone	`/conference/[token]`	Joins conference	🟡
27	Buji AI chatbot	Anyone	Login page → chat with Buji	Responds correctly	🟢
28	Dev routes disabled in production	Dev	GET `/api/v1/dev/token` in production	404 or 403 — not accessible	🔴
29	Health check	Dev	GET `/api/v1/health`	200 with server status	🔴

#	Scenario	Role	Steps	Expected	Phase
30	WhatsApp webhook callback	Auto	Inbound WA message arrives	Processed without 500 error	🟢

DEMO SESSION — FULL FLOW

> Sales creates demo link → prospect joins without login → attends class → takes demo exam → post-demo summary sent.

#	Scenario	Role	Steps	Expected	Phase
31	Sales creates demo link from Sales dashboard	Sales	Sales → Leads → Lead card → Send Demo Link	Demo link generated; WhatsApp message sent to prospect's phone	🔴
32	Prospect opens demo link without login	Guest	Open `/demo/[linkId]` in browser	Demo landing page loads: class name, subject, teacher name shown	🔴
33	Prospect fills name and contact on demo page	Guest	Name, Phone → Join	Proceeds to pre-join lobby	🔴
34	Guest joins live session room	Guest	Enter from lobby	Guest tile appears in room; teacher sees "guest joined" in demo tab	🔴
35	Teacher sees guest in Demo tab	Teacher	Teacher sidebar → Demo tab	Guest name, join time listed	🟡
36	Teacher pushes demo exam to guest	Teacher	Demo tab → Push Exam to Guest	Guest receives exam overlay with MCQ questions	🟡
37	Guest submits demo exam	Guest	Answer questions → Submit before timer	Answers recorded; score shown to guest; result saved	🟡
38	Post-demo summary email sent to sales team	Auto	After demo session ends	Email: guest name, time-in-class, exam score, next-step prompt	🟡
39	AO views demo session history	AO	AO → Demo tab	All demo sessions: date, teacher, guest name, duration, exam score	🟡
40	Teacher views own demo sessions	Teacher	Teacher → Demo tab	Own demo sessions listed with guest details and status	🟡

SALES DASHBOARD — PIPELINE, ACTIVITIES, REMINDERS, REPORTS

#	Scenario	Role	Steps	Expected	Phase
41	Sales creates lead manually	Sales	Sales → Leads → New Lead → name, phone, grade, board, interest	Lead created, status=new; visible in leads list	🔴
42	Sales moves lead through pipeline stages	Sales	Lead → drag to Contacted → Interested → Negotiation → Converted	Stage updates; timestamp recorded in timeline	🟡
43	Sales logs activity against lead	Sales	Lead → Activities → Add → type=call, notes, date	Activity saved; visible in lead timeline	🟡
44	Sales sets follow-up reminder	Sales	Lead → Reminders → Add → due date, note	Reminder created in system	🟡
45	Reminder fires notification on due date	Auto	Wait for reminder due date	Sales sees in-app notification or email alert	🟡
46	Sales views sales reports	Sales	Sales → Reports tab	Conversion rate, leads by stage, demos booked, revenue this month — all shown	🟡

HR DASHBOARD — REMAINING TABS

#	Scenario	Role	Steps	Expected	Phase
47	HR sets fee rate per teacher	HR	HR → Fee Rates tab → select teacher → set rate per session	Rate saved; affects next payslip calculation	🟡
48	HR views cancellation records	HR	HR → Cancellations tab	All cancelled sessions listed: reason, teacher, date, batch	🟡
49	HR creates new HR user (HR Credentials)	HR	HR → HR Credentials → New HR User → email, name	New HR account created; welcome email sent	🟡

AO SETTINGS TAB

#	Scenario	Role	Steps	Expected	Phase
50	AO views platform settings	AO	AO → Settings tab	Settings visible: notification timing, fee defaults, go-live rules, session gate config	🟡
51	AO updates session reminder timing	AO	Settings → reminder_minutes → change 15 → 30 → Save	Setting persisted; next scheduled session uses 30-min reminder window	🟡

Day 13 — 15 May 2026 (Morning)

Theme: EMAIL & WHATSAPP NOTIFICATIONS + FINAL CROSS-BROWSER VERIFICATION

> Test all critical notification types: delivery, content accuracy, links, and timing. Then final cross-browser pass on key flows not fully covered in Days 6–8.

Part 1 — Notification Testing

#	Notification	Trigger	Recipients	Checks	Phase
1	Welcome email — manual enroll	AO enrolls student	Student	Arrives within 2 min, login link works, password correct	🔴
2	Welcome email — parent linked	AO enrolls with parent email	Parent	Arrives, parent login link works	🟡
3	Welcome email — enrollment link	Student self-enrolls + pays	Student	Arrives within 2 min, credentials correct	🔴
4	Invoice email	AO generates invoice	Student	Arrives, `/pay/[token]` link works, correct amount	🔴
5	Receipt email	Student pays invoice	Student + Parent	Both arrive, correct amount + invoice number	🔴
6	Session join link email	AO schedules session (student assigned)	Student	Arrives, join link is correct and functional	🔴
7	Session reminder — WhatsApp (15 min before)	Auto trigger	Student	WA message arrives at 15-min mark, tap-to-join link works	🔴
8	Session reminder — email (15 min before)	Auto trigger	Student	Email arrives at 15-min mark, join link works	🔴
9	Session reminder — parent	Auto trigger	Parent	Parent also receives WA + email reminder	🟡
10	Session cancellation notification	AO cancels session	Student + Parent	Both notified: session name, date, reason	🔴
11	Session reschedule notification	AO edits session time	Student + Parent	Notified with old time and new time	🟡
12	Teacher change notification	AO changes teacher on session	Student + New Teacher	Student gets new teacher info; new teacher sees session	🟡
13	Exam result email	Session ends → auto-grade	Student + Parent	Score, grade letter, session name in email	🔴

#	Notification	Trigger	Recipients	Checks	Phase
14	Payment reminder	AO sends reminder on unpaid invoice	Student + Parent	WA + email delivered with invoice details and payment link	🟡
15	Demo summary email	Demo session ends	Sales team	Post-demo summary: attendees, duration, prospect details	🟢

Part 2 — Final Cross-Browser Verification

#	Scenario	Browser / Device	Expected	Phase
16	Full class flow — Firefox Desktop	Firefox	Teacher + student join, all controls work end-to-end	🔴
17	Full class flow — Edge Desktop	Edge	Same	🔴
18	Full class flow — Safari Mac	Safari Desktop	Same	🔴
19	IE — graceful unsupported page	IE Desktop	"Browser not supported" — no crash, no blank page, helpful message	🔴
20	Screen share — Firefox	Firefox Desktop	Works, prompts screen selection	🔴
21	Screen share — Safari Mac	Safari Desktop	Works with permissions prompt	🟡
22	AI monitoring — Firefox	Firefox Desktop	Active, no errors	🔴
23	AI monitoring — Safari Mac	Safari Desktop	Active or graceful disable	🔴
24	iOS Safari — mic + camera permission	iPhone	Both permission prompts shown; allow → activate	🔴
25	Android Firefox — join + basic controls	Android Firefox	Room loads, mic/cam toggle work	🟡
26	Samsung Internet — join	Samsung browser	Room loads or shows graceful "switch to Chrome"	🟡
27	Role boundary: BC cannot schedule sessions (pending)	BC	No session scheduling option visible in BC nav	🔴
28	Role boundary: BC cannot see other batches	BC	Only assigned batches in list	🔴
29	Role boundary: BC cannot access finance	BC	Finance tab not in nav; direct URL access denied	🔴

#	Scenario	Browser / Device	Expected	Phase
30	Role boundary: student cannot self-register	Student	No signup option on login page; /register returns 404 or redirect	🔴

Day 14 — 15 May 2026 (Afternoon)

Theme: FINAL REGRESSION — Full Flow, All Batch Types, All Devices

> The complete real-life journey from batch creation to class completion.

Full 1:3 Lifecycle (Covers all Days 1–9)

#	Step	Role	Device	Expected
1	AO creates 1:3 batch	AO	Desktop Chrome	Batch created
2	AO checks fee structure	AO	Desktop Chrome	Correct fee for 1:3, Kerala, CBSE shown
3	AO assigns coordinator	AO	Desktop	BC assigned
4	AO manually adds 2 students	AO	Desktop	Both added, welcome emails sent
5	AO creates academic calendar	AO	Desktop Firefox	Calendar created for batch
6	AO adds topics to calendar	AO	Desktop	Chapters/topics listed
7	AO schedules 3 recurring weekly sessions linked to topics	AO	Desktop Firefox	3 sessions on correct dates, topics linked
8	BC assigns teacher to each session	BC	Desktop	Teacher sees all 3 sessions
9	AO generates invoices → students pay → credits added	AO + Students	Desktop	Credits added via Razorpay test
10	AO manually adds 3rd student (mid-series)	AO	Desktop	3rd student gets backfilled tokens for all 3 sessions
11	3rd student pays → gate unlocked	Student 3	Desktop	Credits added
12	Gate test: 0 credits → join	Student 3	iOS Safari	Blocked with payment prompt
13	Restore credits → join	Student 3	iOS Safari	Enters successfully
14	Session reminder arrives 15 min before	All students	WA + email	Delivered at correct time

#	Step	Role	Device	Expected
15	Students open lobby, check device	All 3	Mixed devices	Lobby loads, device check works
16	BC approves teacher go-live	BC	Desktop	Room goes live
17	All 3 students join	S1: Chrome, S2: Android, S3: iOS	Mixed	All present, attendance records created
18	Teacher mic/cam/screen share	Teacher	Desktop	All 3 students see/hear correctly
19	Student 2 raises hand, teacher acknowledges	—	Mixed	Works
20	Chat: all 3 send messages	—	Mixed	All messages delivered
21	Teacher pushes session exam	Teacher	Desktop	Exam sent to all 3
22	All 3 students submit exam	—	Mixed	Auto-graded, grade letters shown
23	Teacher ends class	Teacher	Desktop	All disconnected, session ended
24	Attendance: all 3 present, correct durations	BC	Desktop	Verified
25	Topic marked taught in academic calendar	—	—	Calendar updated
26	Exam result emails received	Students + Parents	—	All 6 emails arrived
27	AO ghost monitors the next session	AO	Desktop	Invisible, sees everything, audit log entry created
28	IE shows graceful error	Student 1	IE Desktop	No crash, helpful message
29	No daily monitoring report to parents	Auto	—	Parent inbox: no daily email
30	Monthly-only report confirmed	—	—	Only monthly report in config

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RUNNING ISSUES TRACKER

🔴 Red Zone — Open

Bug ID	Feature	Description	Assigned	Status
—	—	—	—	—

🟡 Yellow Zone — Open

Bug ID	Feature	Description	Assigned	Status
—	—	—	—	—

🟢 Green Zone — Open

Bug ID	Feature	Description	Assigned	Status
—	—	—	—	—

✅ Resolved

Bug ID	Feature	Fix Summary	Fixed On	Verified By
—	—	—	—	—

DAILY SIGN-OFF CHECKLIST

```
Date          : ___________
Tester(s)     : ___________
Batch Types   : [ ] 1:1   [ ] 1:3   [ ] 1:15   [ ] 1:30   [ ] 1:many
Devices       : [ ] Desktop Chrome   [ ] Desktop Safari   [ ] Desktop Firefox
                [ ] Desktop Edge     [ ] IE               [ ] Android Chrome
                [ ] Android Firefox  [ ] iOS Safari       [ ] iOS Chrome
Bugs Found    : ___   🔴 Critical: ___   🟡 Important: ___   🟢 Polish: ___
Bugs Closed   : ___
Blocked Tests : ___
Overall Status: 🔴 / 🟡 / 🟢
Notes         : ___________
Sign-off      : ___________
```

GREEN LAUNCH CRITERIA — 15 May 2026

Platform Setup

- ☐ All batch types create correctly (1:1, 1:3, 1:15, 1:30, lecture)
- ☐ Fee structures correct for all region/board/batch-type combinations
- ☐ AO assigns coordinators; BC sees only assigned batches
- ☐ AO schedules sessions (BC session scheduling pending implementation); BC cannot create batches or access finance

Student Acquisition & Enrollment

- ☐ AO manually adds student — account created, welcome email delivered, student can login
- ☐ Enrollment link → student self-signup → Razorpay payment → account created

- ☐ AO assigns enrolled student to batch → join tokens generated
- ☐ Batch created AFTER student enrolls — AO creates batch to house the student group
- ☐ Invoice → Razorpay → credits added end-to-end
- ☐ Public payment page `/pay/[id]` works without login
- ☐ Session gate blocks zero-credit students with payment prompt
- ☐ Demo link (secondary flow) → WhatsApp → prospect opens on mobile → guest joins as view-only
- ☐ CRM students (smartuplearningventures@gmail.com) verifiable by AO

Academic Calendar & Scheduling

- ☐ Academic calendar created, topics mapped to sessions
- ☐ Full week scheduled: 5 sessions across different subjects
- ☐ Recurring sessions on correct dates
- ☐ New student added mid-batch → backfill tokens → joins successfully
- ☐ BC assigns teachers per session (including multiple teachers per subject)

Pre-Class

- ☐ Session reminder (WhatsApp + email) arrives exactly 15 min before
- ☐ Pre-join lobby loads; handles mic/cam denied and no camera gracefully
- ☐ BC go-live approval flow works (and bypass toggle works)

Live Class

- ☐ Audio and video stable on Desktop, Android, iOS
- ☐ Screen share works on Chrome, Firefox, Safari
- ☐ All batch types stable (1:30 and lecture with 30/50 students)
- ☐ IE shows graceful "browser not supported" — no crash
- ☐ Teacher controls (mute, remove, spotlight, whiteboard, recording) all work
- ☐ Session exam push → student answers → auto-grade → result email

Post-Class & Ongoing

- ☐ Attendance correct: present/late/absent/duration for all batch sizes
- ☐ Block/unblock student → gate applies correctly
- ☐ Deactivate teacher → sessions flagged → substitute assignable
- ☐ No daily parent AI report — monthly only confirmed
- ☐ Ghost observer is fully invisible; ghost_audit_log entry created

Other Portals

- ☐ Owner custom permissions override role defaults
- ☐ HR approves teacher leave → sessions flagged
- ☐ Sales can send demo links
- ☐ Parent views attendance, fees, exam results

Final

- ☐ All 14 daily sign-offs complete — zero open 🔴 bugs
- ☐ 420 test cases passed or documented
- ☐ Health check `/api/v1/health` returns 200

- ☐ Dev routes (`/dev/token`, `/dev/livekit-test`) inaccessible in production

Document version: 4 (rewritten 28 April 2026) Owner: Dev Team + Batch Coordinators Review: CEO, Academic Operator Deadline: 15 May 2026 | Total: 420 test cases